RAJALAKSHMI ENGINEERING COLLEGE

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D2

Question 1 | Correct Marked out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main()
{
    int n;
    scanf("%d", &n);

    int c = 0;
    int count = 1;   // c = 0

    int i = n / 2;

    while (1)
    {
        count++;     // outer loop condition: i < n
        if (i >= n)
            break;

        int j = 1;

        while (1)
        {
            count++;     // middle loop condition: j < n
            if (j >= n)
                break;

            int k = 1;

            while (1)
            {
                count++;     // inner loop condition: k < n
                if (k >= n)
                    break;

                c++;
                count++;     // c++

                k = k * 2;
            }

            j = j * 2;
        }

        i++;
    }
```

```
46
47      printf("%d", count);
48
49      return 0;
50  }
```

Check

	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

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